



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VENKATESHWARA FUELS UNIT II (CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

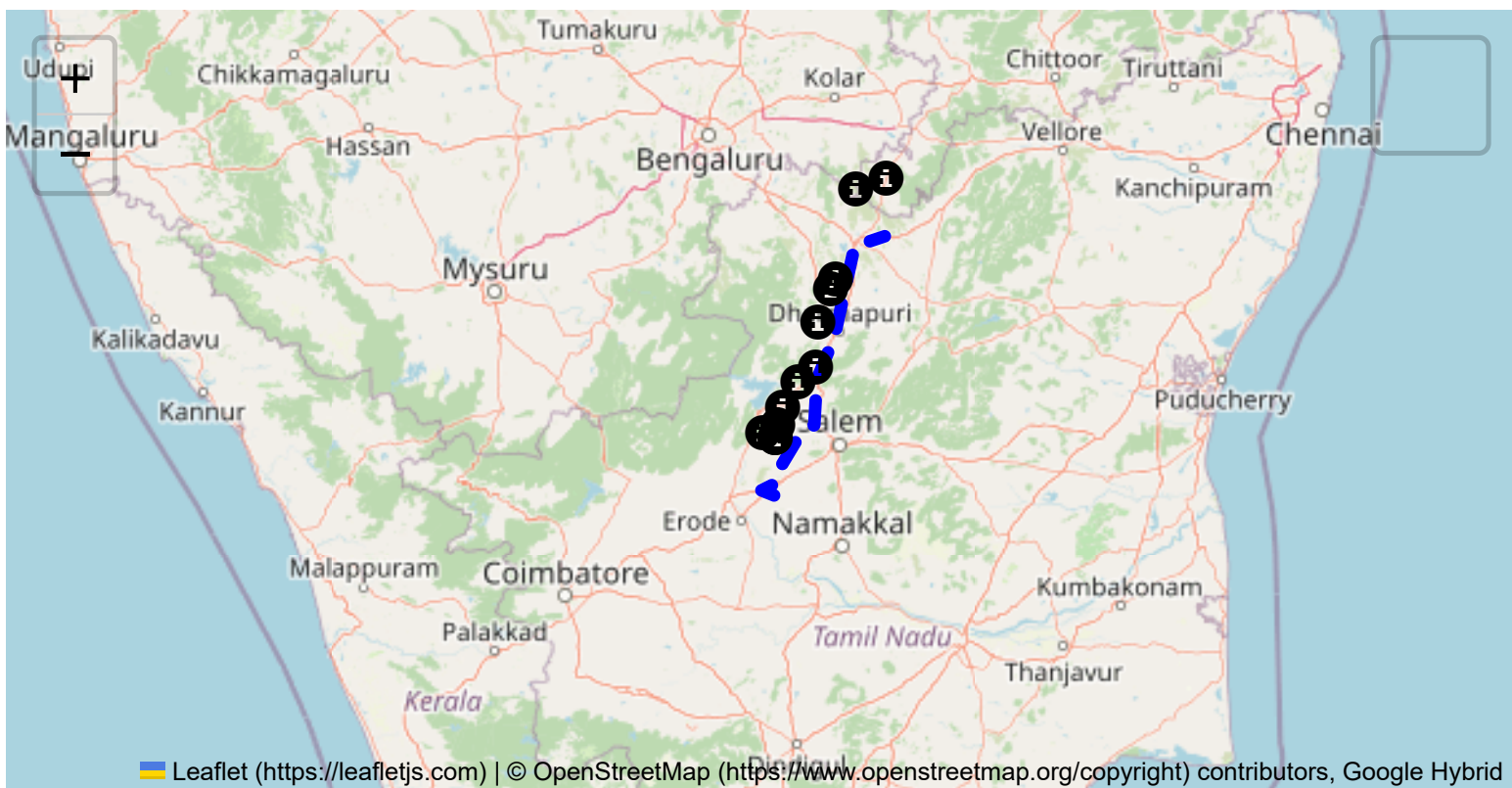
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

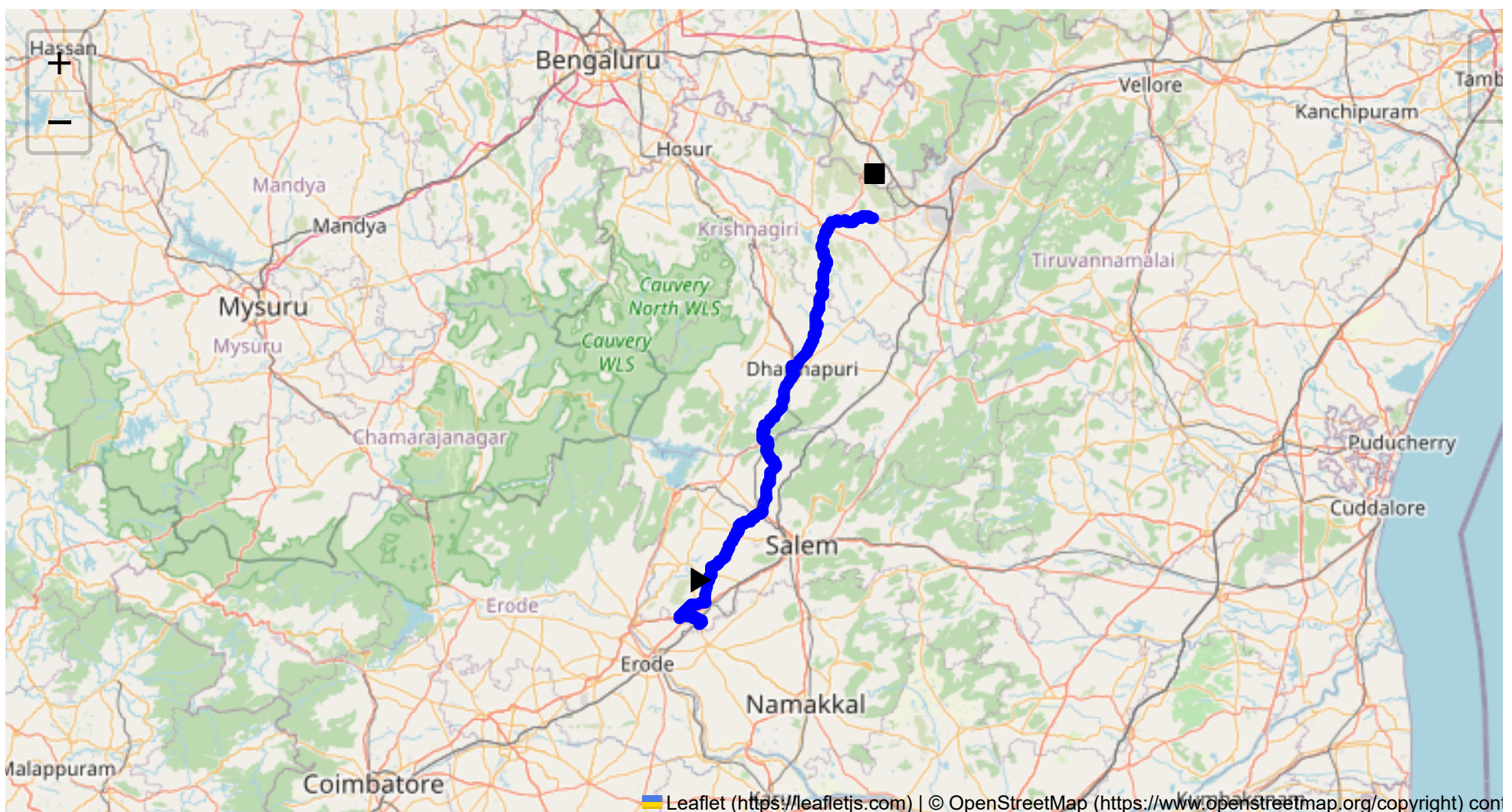
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 167.40 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 4.0 hours
Start: (11.4381, 77.8734)
End: (12.5437379, 78.3545254)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Bargur through the specified waypoints spans around 167.40 kilometers. Starting from CVQF+23W in Sangagiri and ending at G9V3+FWW in Bargur, it traverses multiple towns and different road types, primarily including highways and state roads. The journey involves traveling along

main highways such as the Salem-Kochi Highway and SH 86, offering a generally straightforward path conducive for heavy vehicle movement.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The climate in Tamil Nadu is generally tropical, with high humidity, especially during the summer months (March to May). The monsoon season (June to September) can bring heavy rainfall, leading to potential flooding and road surface issues, particularly in rural or poorly drained sections. Travelers should be cautious of reduced visibility and slippery conditions during these months.

3. Analysis of Traffic Patterns

Traffic congestion peaks typically occur during the morning (7:30 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) along major highways near urban centers like Salem and Krishnagiri. Roads leading into and out of these towns may experience bottlenecks. It's advisable to schedule trips during off-peak hours to avoid delays.

4. Assessment of Road Quality and Infrastructure

The highways, such as the Salem-Kochi Highway and SH 86, are generally well-maintained, facilitating smooth transit for heavy vehicles. However, there can be variable road conditions in rural or less frequently maintained sections, so drivers should be vigilant for potholes especially post-monsoon.

5. Suggestions for Alternative Routes for Emergencies

In the event of disruption, consider alternate paths via NH 44 and NH 179A, connectivity permitting. These national highways can offer quicker detours but analyzing current traffic and road conditions is crucial.

6. Summary of Local Regulations Affecting Hazardous Material Transport

In Tamil Nadu, transporting hazardous materials requires strict adherence to the Central Motor Vehicle Rules. Ensure appropriate placarding, proper documentation, and compliance with permitted travel hours. Special restrictions during major public holidays may apply.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Although major incidents involving hazardous materials aren't extensively documented publicly, typical issues like vehicular accidents due to overloading or driver negligence can occur. Staying informed via local news channels or official advisories helps anticipate areas with recurrent incidents.

8. Environmental Considerations and Sensitive Areas

The route passes near environmentally sensitive zones, particularly around certain forested sections. Drivers should strictly adhere to speed limits and avoid stopping in unauthorized areas to protect local

wildlife.

9. Analysis of Communication Coverage

Generally, mobile coverage is reliable in urban and suburban stretches but may weaken in remote rural areas or forested sections, such as near Thoppur. Utilizing truck-specific communication devices that use broader signal bands is recommended.

10. Estimated Emergency Response Times for Different Route Segments

Urban areas such as Salem and Krishnagiri enjoy quicker emergency response times (approximately 15-30 minutes). Rural stretches may experience delayed responses, often 45 minutes to an hour, depending on proximity to major towns.

11. An Overall Summary of Risk Assessment

The route presents moderate risk, primarily due to variable traffic conditions and the potential for weather disruption during monsoon months. Adhering to best practices in transportation safety and planning journeys outside peak hours can mitigate most risks. Ensuring compliance with safety regulations for transporting hazardous materials is vital, as is staying prepared for communication and emergency response limitations in rural areas. Regularly updated navigation tools can aid in avoiding congestion and discovering alternate routes swiftly if necessary.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45545, 77.81812	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56352, 77.90078	30 KM/Hr
11	Turn	Medium	11.59189, 77.90645	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.59328, 77.90809	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	High	11.64217, 77.95200	15 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	Medium	11.67858, 77.96834	30 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72504, 78.01680	30 KM/Hr
21	Turn	Medium	11.73766, 78.03445	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74120, 78.04588	30 KM/Hr
24	Turn	Medium	11.86805, 78.08353	30 KM/Hr
25	Turn	Medium	11.87913, 78.08206	30 KM/Hr
26	Turn	Medium	11.92566, 78.06503	30 KM/Hr
27	Turn	Medium	11.93307, 78.06426	30 KM/Hr
28	Turn	Medium	11.93545, 78.06130	30 KM/Hr
29	Turn	Medium	11.93694, 78.05425	30 KM/Hr
30	Turn	Medium	11.95531, 78.04698	30 KM/Hr
31	Turn	High	11.95803, 78.04964	15 KM/Hr
32	Turn	Medium	11.96068, 78.04960	30 KM/Hr
33	Turn	High	11.96370, 78.05142	15 KM/Hr
34	Turn	High	11.96542, 78.04938	15 KM/Hr
35	Turn	Medium	11.96650, 78.04927	30 KM/Hr
36	Turn	Medium	11.96890, 78.05109	30 KM/Hr
37	Turn	Medium	12.12094, 78.13269	30 KM/Hr
38	Turn	Medium	12.14025, 78.12819	30 KM/Hr
39	Turn	High	12.14099, 78.12862	15 KM/Hr
40	Turn	High	12.14056, 78.13957	15 KM/Hr
41	Turn	High	12.24611, 78.19519	15 KM/Hr
42	Turn	Medium	12.25200, 78.20034	30 KM/Hr
43	Turn	High	12.40026, 78.21775	15 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	Medium	12.50271, 78.22341	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	High	12.50709, 78.22230	15 KM/Hr
47	Turn	Medium	12.52414, 78.23199	30 KM/Hr
48	Turn	Medium	12.53197, 78.24014	30 KM/Hr
49	Turn	Medium	12.54851, 78.33415	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
23	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
24	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
26	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
29	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	hospital	Government Hospital, Bargur	12.5444, 78.357225	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
25	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
27	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
28	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
30	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
31	college	Government Boys Arts and Science college	12.5323457, 78.2457274	30 km/h	Medium
32	college	Government Polytechnic College	12.5352064, 78.2460463	30 km/h	Medium
33	college	IRT Polytechnique College	12.5354149, 78.2989061	30 km/h	Medium
34	college	P. S. V. College of Engineering and Technology	12.5350134, 78.3023583	30 km/h	Medium
35	school	Kingsley Garden Matriculation Higher Secondary School	12.5450463, 78.3151896	30 km/h	Medium
36	college	Government Engineering College	12.5442249, 78.331078	30 km/h	Medium
38	school	Goverment Girls Higher Sec School	12.5440902, 78.3581379	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56352, 77.90078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93307, 78.06426



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93545, 78.06130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95803, 78.04964



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96068, 78.04960



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96890, 78.05109



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.14025, 78.12819



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14099, 78.12862



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25200, 78.20034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



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Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50271, 78.22341



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

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Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52414, 78.23199



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.53197, 78.24014



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.54851, 78.33415
